Retrieval on Old Dutch Documents

Yiming Xu, Gerard Asbert, Martin Kostelník, Koen van Schuijlenburg, Jan-Frederik Stock

Problem Statement

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Information retrieval in historical documents:

- Spelling, grammar, and vocabulary mismatch between old and modern language
- Embeddings of the historical text and queries must match
- Solution: Training a model to translate queries to the old language, or the historical text to the modern language

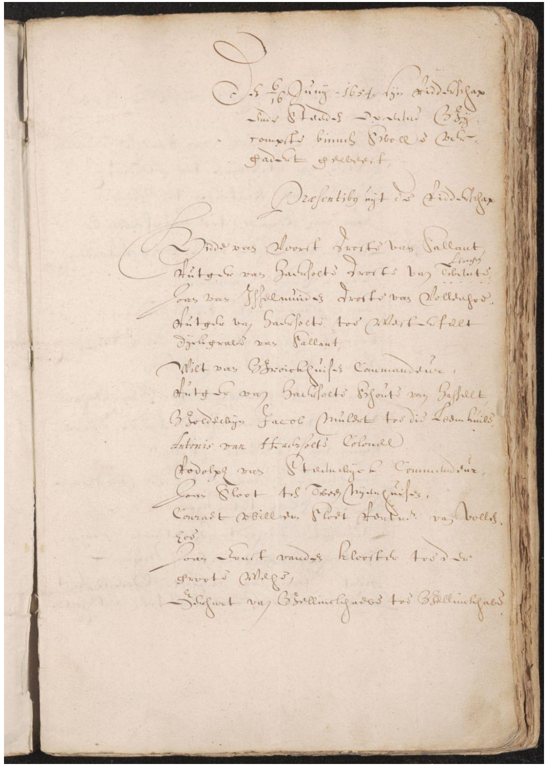
This results in **added difficulty** to make reliable models for any task.

Proposed Method

For this project, we chose **resolutions** written in **Old Dutch** in **1550** as our document type. **Why?**

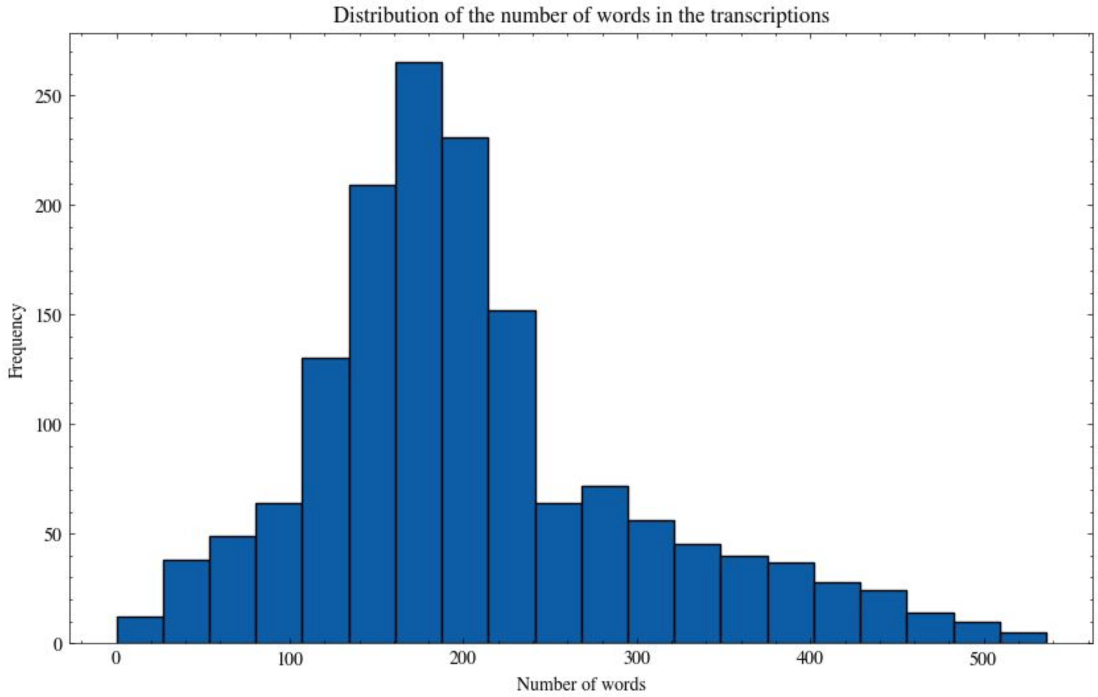
- Lack of annotated data.
- Large models majorly haven't seen data in Old Dutch.
- Inconsistent spelling variations.

This results in added difficulty to make reliable models for any task.



Dataset

A subset of the dataset contains **1545** pages in historical Dutch.



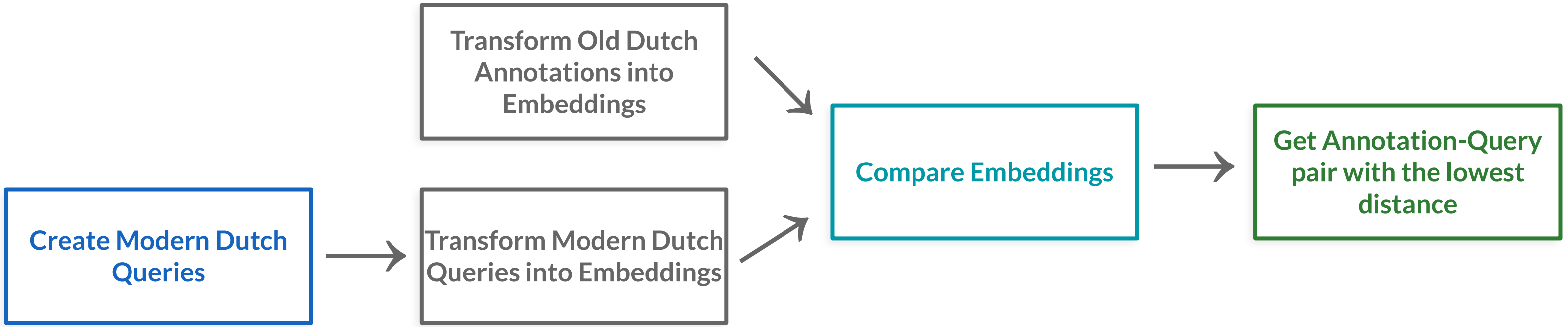
Baselines

1. Translate queries from modern Dutch to old Dutch
2. Translate the collection from old to modern Dutch

Task

For the task we chose **Retrieval**.

In this case, the normal way retrieval is done is via the following pipeline:



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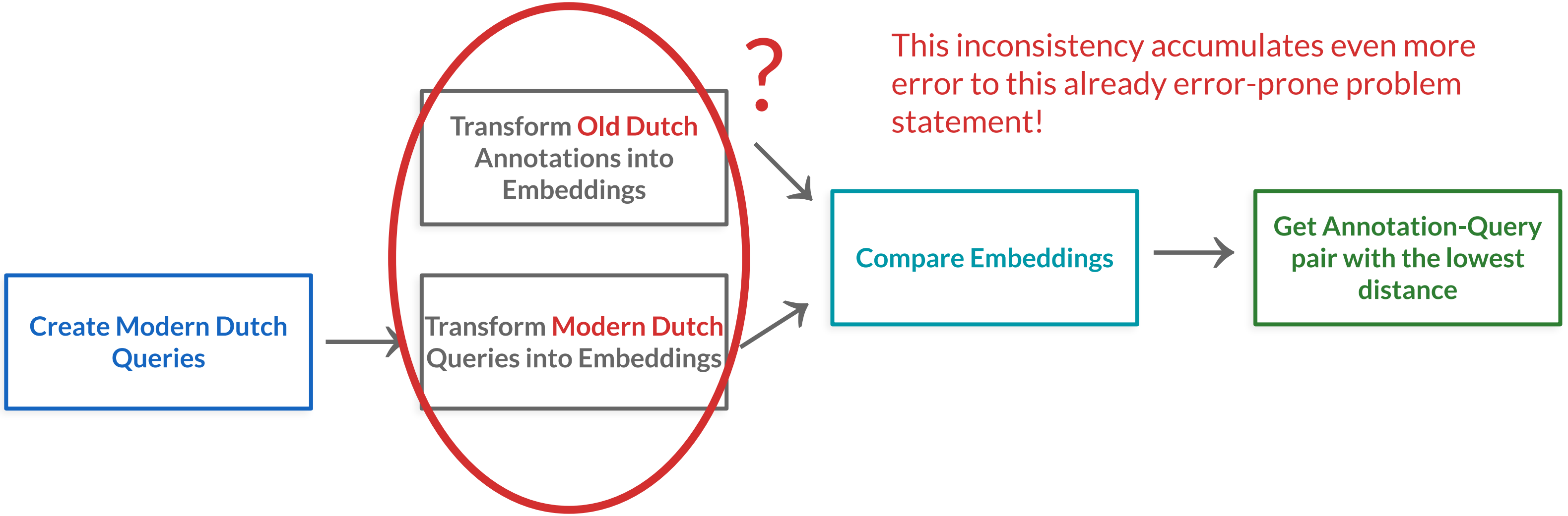

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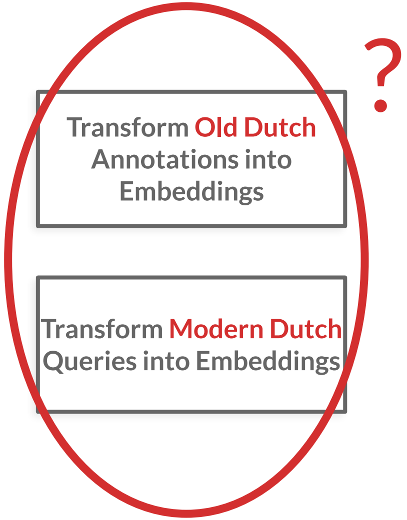

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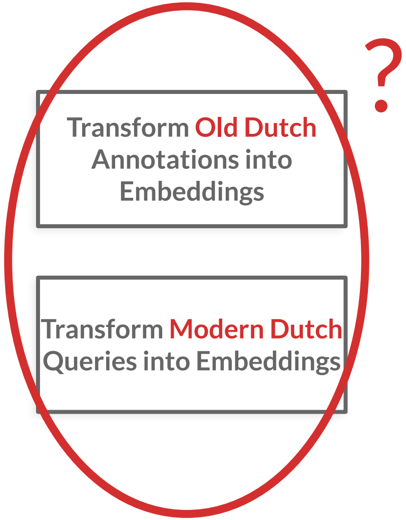

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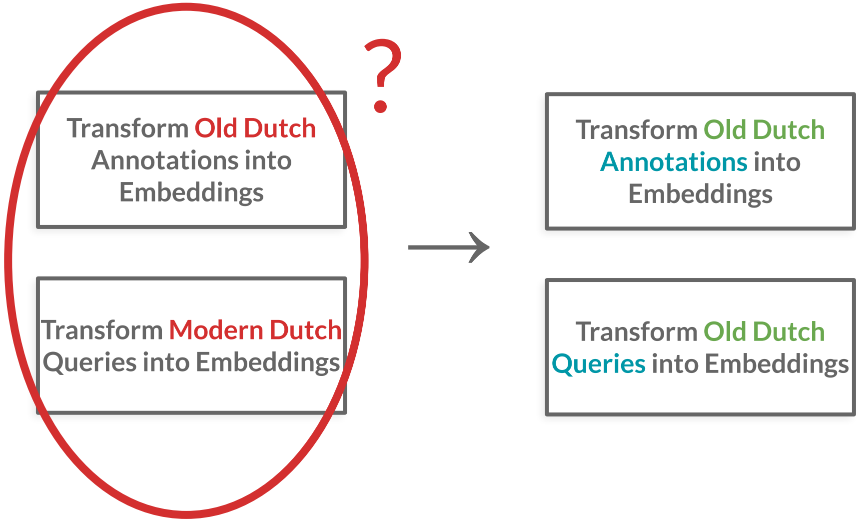

- We don’t modify the groundtruth annotations.
- Avoid artifacts in GT.

Task

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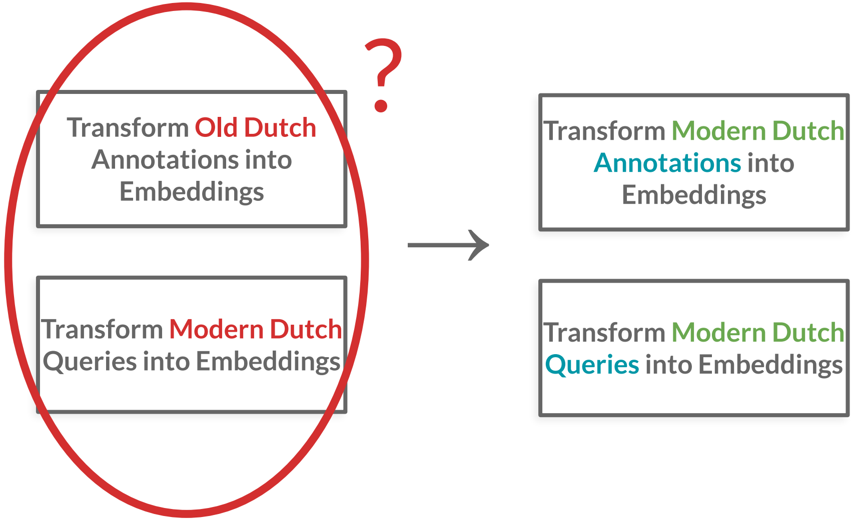
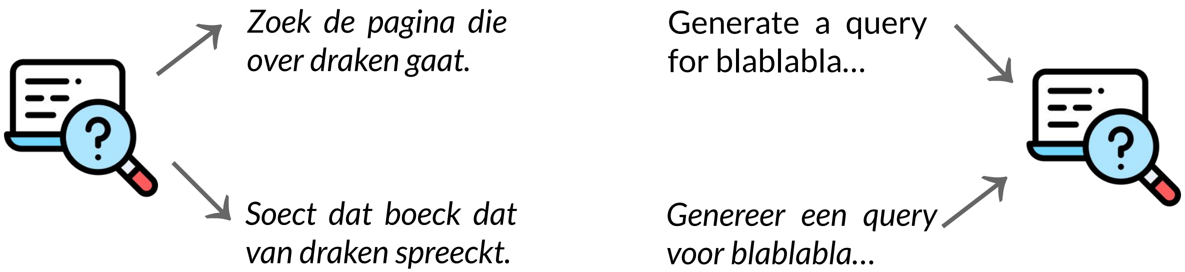


- Most modern embedding models are more used to Modern Dutch
- Easier debugging (Coen is dutch :P)

Methodology

Query generation

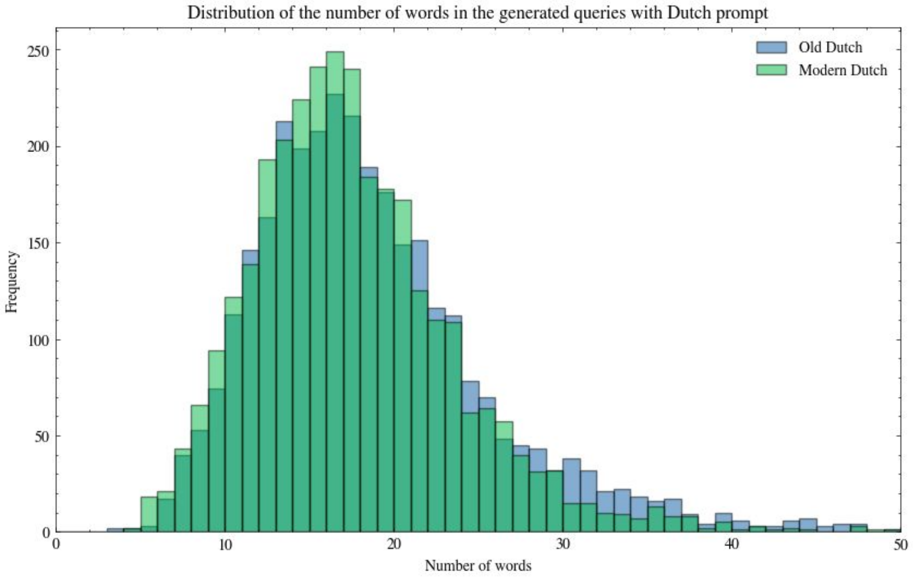
For the query generation we used the **Llama3.1 8B** model, given its a fairly potent model while **fitting in a single GPU.**

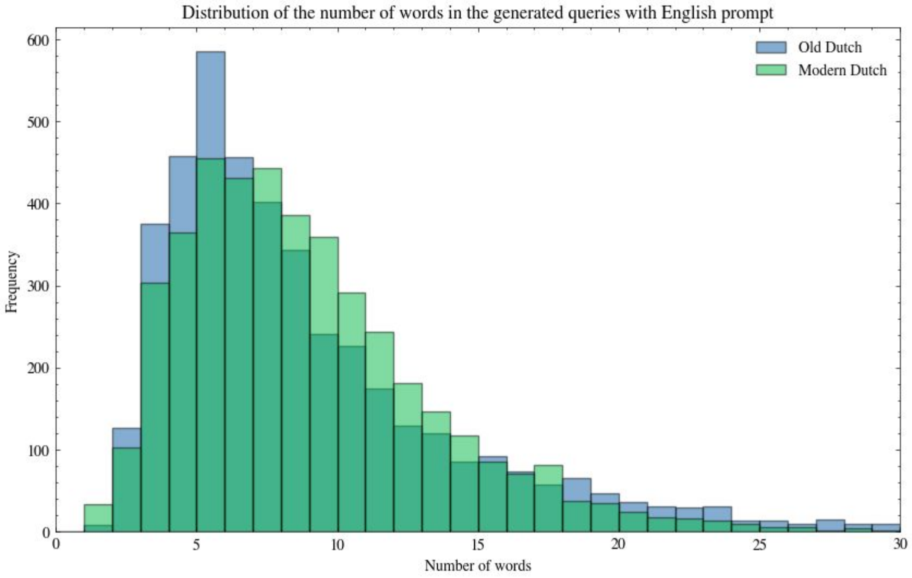


Sometimes the LLM **failed** to generate the queries:

English prompt – (137/1545) -> (8.86 %)
Dutch prompt – (537/1545) -> (34.76 %)

Query generation – Lengths





Embedding Generation

For the embedding generation we used **five different models**, each one with its own specifications:

BM25

- Exact word overlap
- Very weak semantic understanding.

BGE-M3

- Handles cross-lingual retrieval well
- Not trained specifically on Old Dutch

Qwen3-0.6B

- Not a dedicated embedding model
- But strong semantic reasoning for its size

mE5-large

- Strong multilingual embeddings
- Still not trained on Old Dutch or any other ancient languages

E5-NL-large

- Dutch-specialized
- Not trained in Old Dutch though

Results

Modern Queries vs Old Queries

Table 2: Overall retrieval performance on queries generated from the English prompt. Best per column in **bold**.

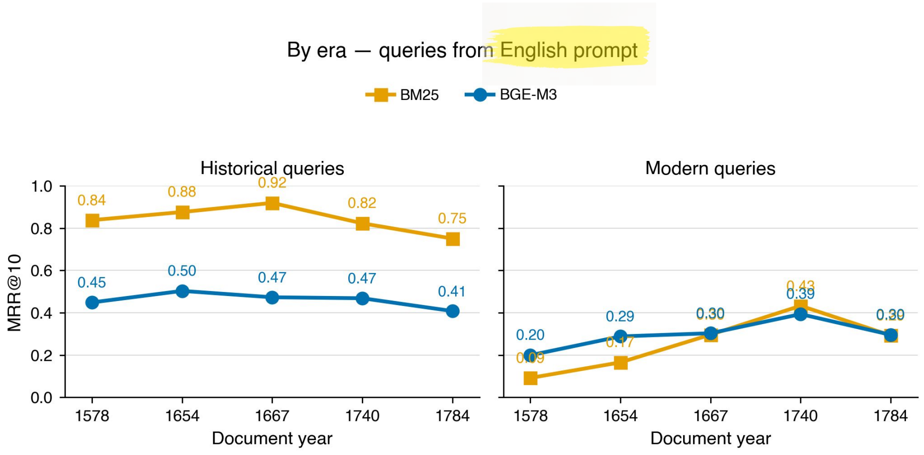
	Historical query			Modern Dutch query		
Model	R@1	R@10	MRR	R@1	R@10	MRR
BM25	**0.797**	**0.924**	**0.844**	0.155	0.310	0.202
BGE-M3	0.377	0.639	0.457	**0.199**	**0.429**	**0.263**
Qwen3-0.6B	0.312	0.578	0.393	0.175	0.395	0.240
mE5-large	0.254	0.521	0.333	0.169	0.387	0.231
E5-NL-large	0.210	0.457	0.281	0.144	0.356	0.202

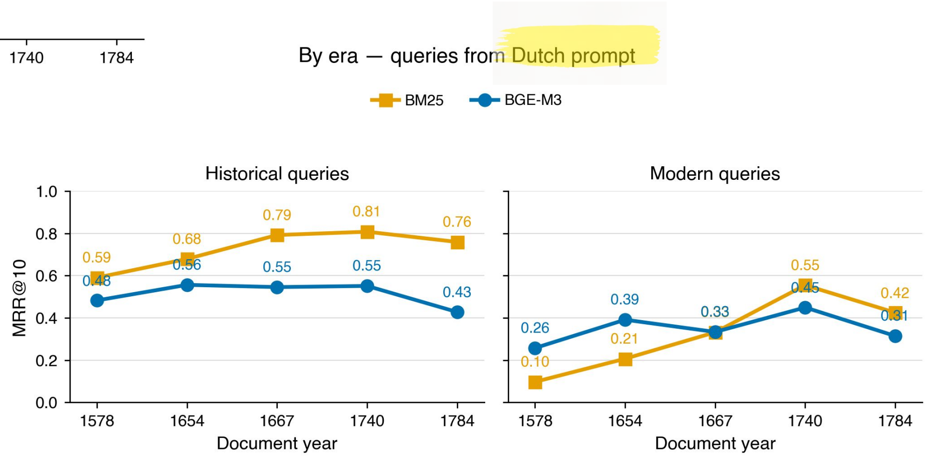
Modern Queries vs Old Queries

Table 1: Overall retrieval performance on queries generated from the Dutch prompt. Best per column in **bold**.

	Historical query			Modern Dutch query		
Model	R@1	R@10	MRR	R@1	R@10	MRR
BM25	**0.647**	**0.752**	**0.681**	0.197	0.335	0.240
BGE-M3	0.427	0.680	0.504	0.242	**0.515**	0.315
Qwen3-0.6B	0.383	0.636	0.461	0.241	0.509	0.320
mE5-large	0.318	0.589	0.400	**0.247**	0.514	**0.327**
E5-NL-large	0.249	0.520	0.330	0.181	0.431	0.254

Analysis by Era





Modern Queries vs Old Queries

Table 3: MRR@10 of each retriever on the same corpus under two query-generation prompts (English vs. Dutch). Best per column in **bold**. Switching the LLM prompt to Dutch produces more paraphrased queries, narrowing BM25's advantage on historical queries and lifting all dense models.

	Historical Dutch query		Modern Dutch query	
Prompt	English	Dutch	English	Dutch
BM25	**0.844**	**0.681**	0.202	0.240
BGE-M3	0.457	0.504	**0.263**	0.315
Qwen3-0.6B	0.393	0.461	0.240	0.320
mE5-large	0.333	0.400	0.231	**0.327**
E5-NL-large	0.281	0.330	0.202	0.254

Modern Annotations vs Old Annotations

Table 4: Retrieval performance on a modern-Dutch corpus of 69 page-level translations. Best per column in **bold**.

	Historical Dutch query			Modern Dutch query		
Model	R@1	R@10	MRR	R@1	R@10	MRR
BM25	0.411	0.809	0.540	0.340	0.766	0.478
Qwen3-0.6B	0.426	0.773	0.538	0.433	0.830	0.559
mE5-large	0.468	0.794	0.566	0.433	**0.901**	0.579
E5-NL-large	0.397	0.809	0.520	0.418	0.872	0.561
BGE-M3	**0.475**	**0.879**	**0.608**	**0.475**	0.865	**0.600**

Conclusions

Conclusions

- **Pu** sabotaged us **multiple times.**
- Old Dutch queries **performed better** than just using Modern Dutch queries
- We're gonna be **rich** (800€ specifically)

Future work

Self-supervised translating historic to modern Dutch

- Use a seq2seq model to translate old Dutch to modern Dutch
- Correct the grammar of the translated text
- Use a seq2seq model to translate the corrected text back to old Dutch